

RADAR

Real-Time Vehicle Analysis System

Egyptian License Plate Detection · OCR · Seatbelt & Mobile Detection

YOLOv11m

YOLO26m

YOLO26s

Streamlit

LapSRN

Real-ESRGAN

The Problem



Traffic Enforcement



Manual monitoring of vehicles, license plates, and driver behavior is slow, error-prone, and requires significant human resources.



License Plate OCR



Egyptian plates have unique Arabic characters and dual-language format (Arabic + numbers). Traditional OCR struggles with this complexity.



Driver Safety



Seatbelt violations and mobile phone usage while driving are leading causes of accidents. Manual detection is inconsistent.



Real-Time Processing

Traffic systems need fast, automated analysis that can process live feeds and images efficiently.

01

System Architecture

How RADAR processes images end-to-end

End-to-End Pipeline



Roboflow-Style Live Filtering

All models run ONCE at minimum confidence (0.01) and cache results. Sidebar sliders filter cached results client-side — no model re-run on slider change. This gives real-time interactive filtering without GPU overhead.

02

Our Models

Six specialized models working together

Model Arsenal

Plate Detector

YOLOv11m

1 classes

Locate license plates in images

OCR V1

YOLOv11m

38 classes

Character recognition (baseline)

OCR V2

YOLO26m

38 classes

Improved character recognition

OCR V2 Weighted

YOLO26m

38 classes

Weighted loss for better precision

Car Detector

YOLO26s

3 classes

Vehicle detection (car/bus/truck)

Seatbelt+Mobile

YOLOv11m

5 classes

Driver safety violation detection

03

Plate Detection

YOLOv11m — Finding license plates in the wild

License Plate Detection

YOLOv11m | Fine-tuned on Egyptian car plates

Architecture	YOLOv11m (medium)
Input Size	640 × 640 pixels
Detection Target	Egyptian license plates
Confidence Threshold	0.25
Output	Bounding box + confidence
Post-processing	Sorted widest-first for best OCR

Why Plate-First Detection?

- ▶ Detects plates in the full image before cropping — avoids missing small plates
- ▶ Plates sorted by width (widest first) — best quality crop for OCR
- ▶ Runs at cache-min confidence (0.01) then filtered client-side at 0.25
- ▶ Single model handles all Egyptian plate styles (white, blue, old format)

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OCR Models

Reading characters from Egyptian license plates

The OCR Challenge

Egyptian plates have unique challenges



Dual Language



Plates contain both Arabic letters and Latin/Arabic numerals — OCR must handle bidirectional text



38 Character Classes



10 digits + 28 Arabic letters with visually similar characters (e.g., ع vs ص, غ vs ض)



Low Resolution



Cropped plates from far vehicles are blurry — super-resolution enhancement is critical



Varied Conditions



Different lighting, angles, plate styles (white, blue), and degradation levels

OCR V2 — Training Results

YOLO26m | 100 epochs | batch=16 | AdamW | characters_final dataset (9,324 images)

99.2%

mAP@50

77.4%

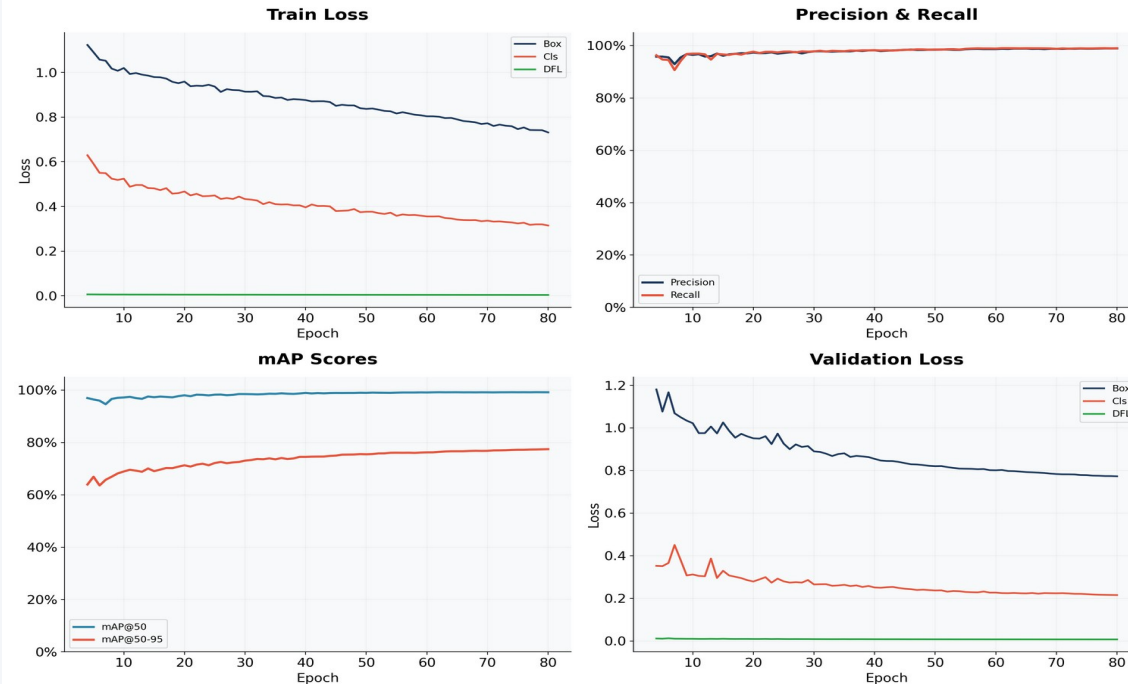
mAP@50-95

98.9%

Precision

99.0%

Recall

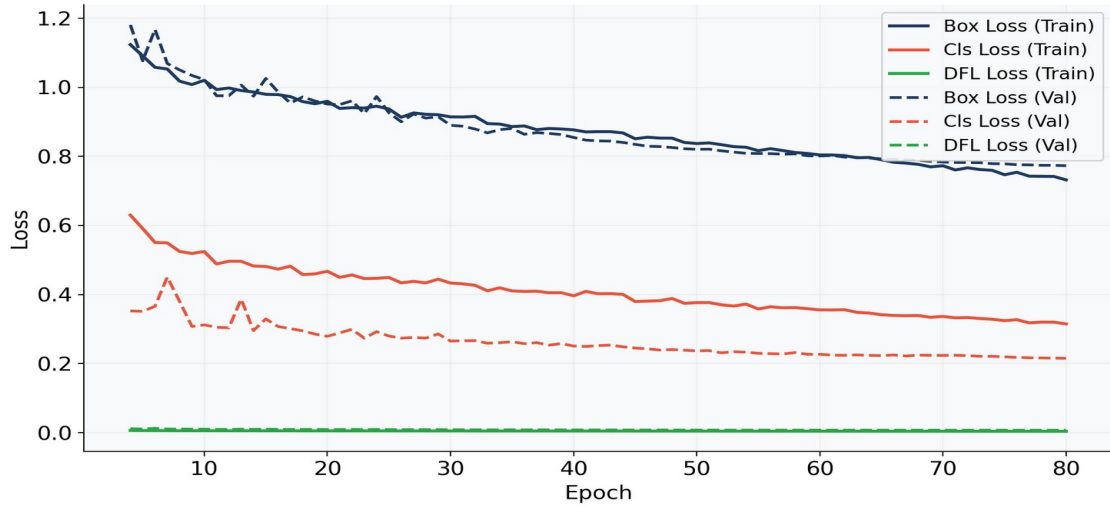


Training Configuration

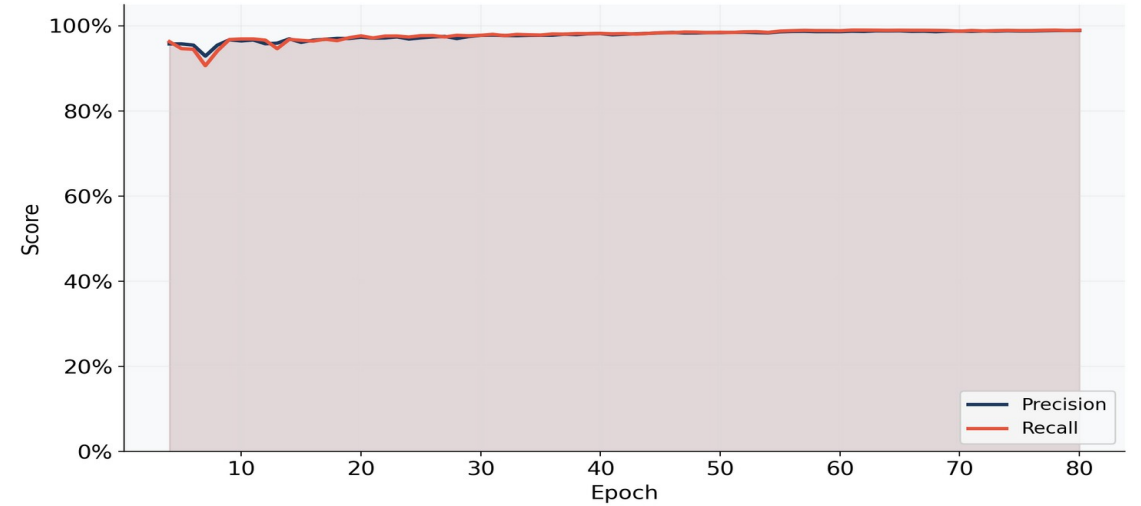
- Architecture: YOLO26m (medium)
- Dataset: characters_final — 9,324 images
- Image Size: 640 × 640
- Optimizer: AdamW, lr=0.01
- Augmentation: RandAugment + MixUp(0.1)
- Warmup: 3 epochs (removed from charts)
- Best epoch: 78 / 100
- Platform: Kaggle T4 GPU

OCR V2 — Training Curves

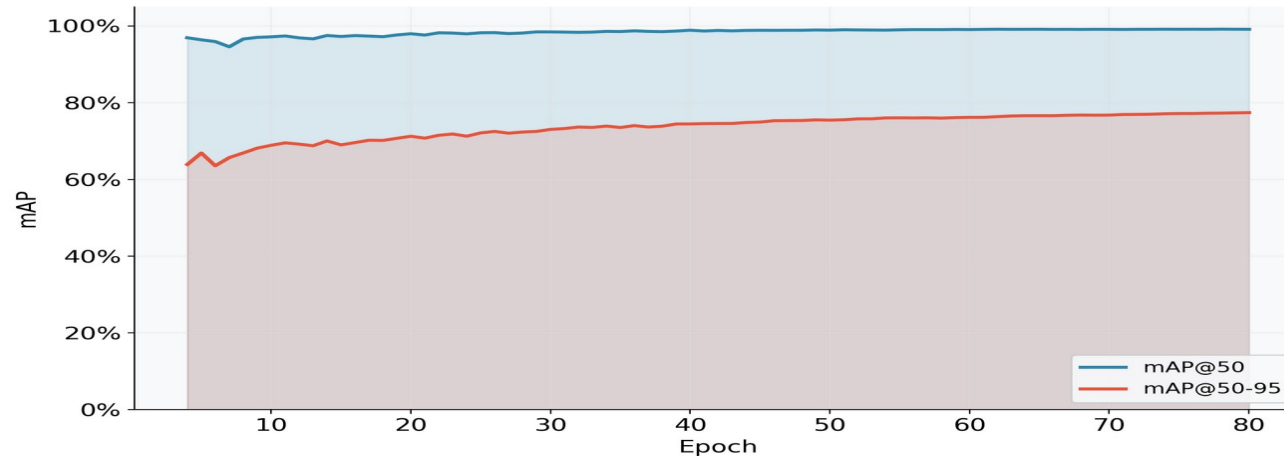
OCR V2 (YOLO26m) — Training & Validation Loss



OCR V2 (YOLO26m) — Precision & Recall



OCR V2 (YOLO26m) — mAP Score



OCR Model Comparison

Three OCR models run side-by-side for best accuracy

OCR V1

YOLOv11m

Baseline character model

Training Data

Egyptian plates dataset

Each model runs on 3 versions:

- Original crop
- LapSRN 2× enhanced
- Real-ESRGAN enhanced

OCR V2

YOLO26m

Improved with characters_final dataset

Training Data

9,324 images, 100 epochs

Each model runs on 3 versions:

- Original crop
- LapSRN 2× enhanced
- Real-ESRGAN enhanced

OCR V2 Weighted-3

YOLO26m

Weighted BCE loss for class balance

Training Data

Same data, weighted training

Each model runs on 3 versions:

- Original crop
- LapSRN 2× enhanced
- Real-ESRGAN enhanced



Total: 3 models × 3 enhancements = 9 OCR results per plate for maximum accuracy

05

Super Resolution

AI-powered image enhancement for better OCR

AI Super-Resolution Pipeline

1

Original Crop

Raw plate crop from plate detector — may be blurry or pixelated, especially for distant vehicles

Baseline — no processing applied

2

LapSRN 2×

Laplacian Pyramid Super-Resolution Network — upscales 2× using deep learning. Fast inference, good for mild blur.

LapSRN_x2.pb — can crash on certain shapes, wrapped in try/except fallback

3

Real-ESRGAN

Real-World Super-Resolution GAN — handles complex degradations (blur, noise, compression artifacts). Higher quality but slower.

Excels at recovering text details from degraded plate images

06

Seatbelt & Mobile Detection

Driver safety violation detection

Seatbelt & Mobile Detection

YOLOv11m | 80 epochs | batch=32 | 6,424 images | 5 classes

Detection Classes

	person-noseatbelt	Driver/passenger without seatbelt
	person-seatbelt	Driver/passenger wearing seatbelt
	seatbelt	Seatbelt strap visible
	windshield	Windshield detection (context)
	mobile	Mobile phone in use

Dataset Construction

- ▶ Merged from 3 sources: v3 base + v2 mobile + v5 unique
- ▶ 6,424 images at 640 × 640 resolution
- ▶ 484 mobile phone labels
- ▶ Train/Valid/Test split with balanced distribution
- ▶ Built from v3 base (NOT v4 which was 320×320)

90.0%

mAP@50

54.3%

mAP@50-95

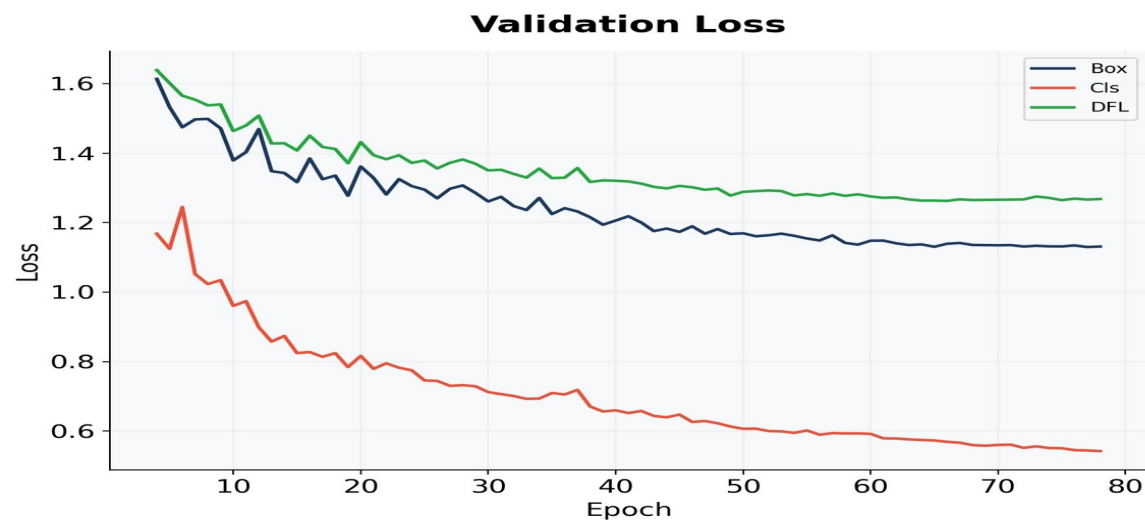
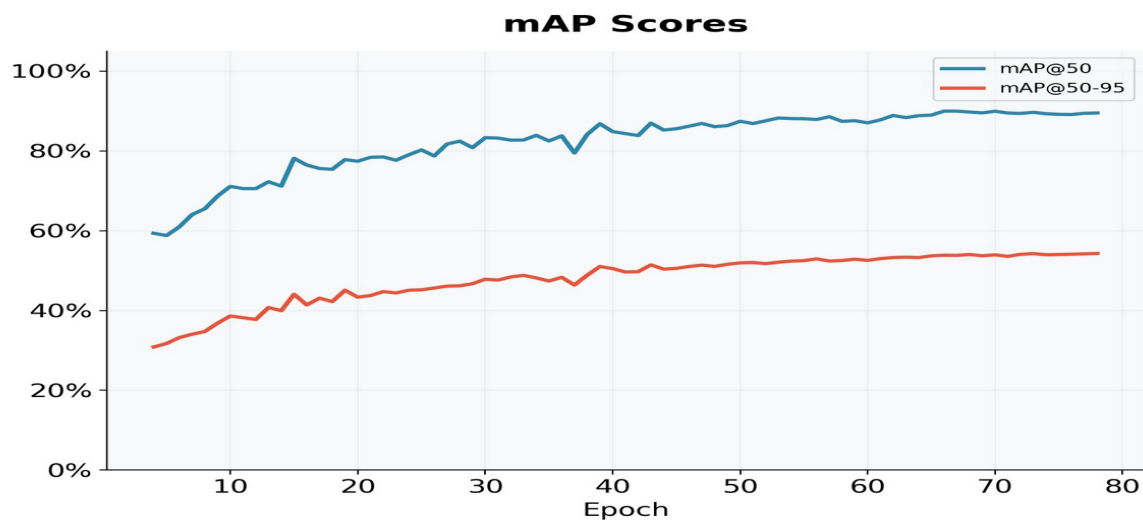
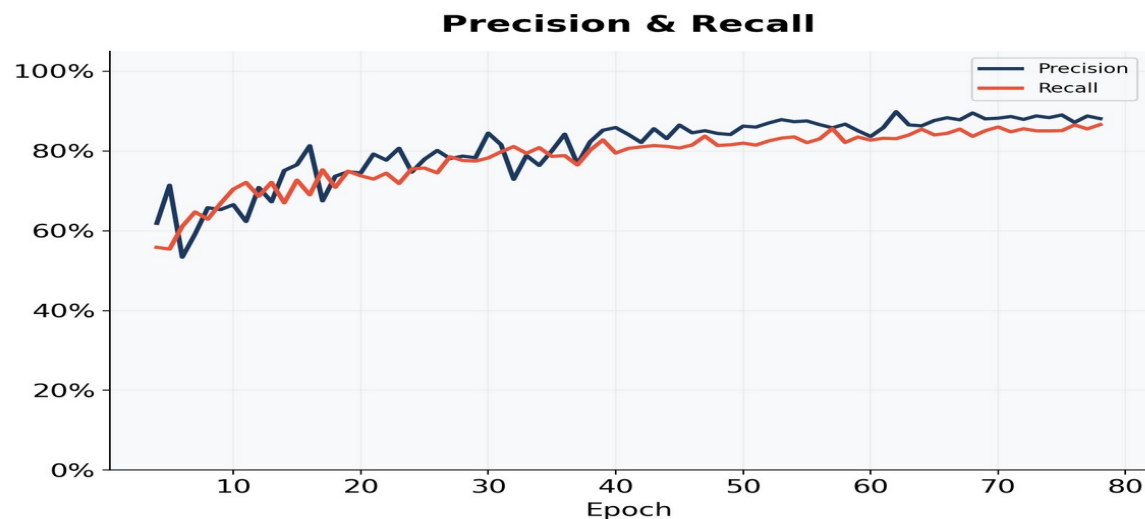
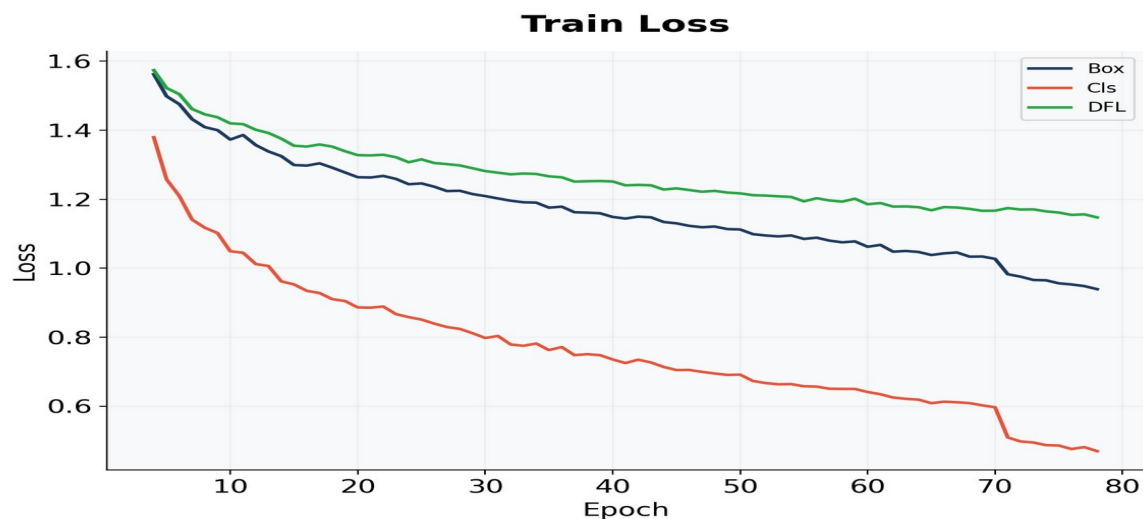
89.8%

Precision

86.6%

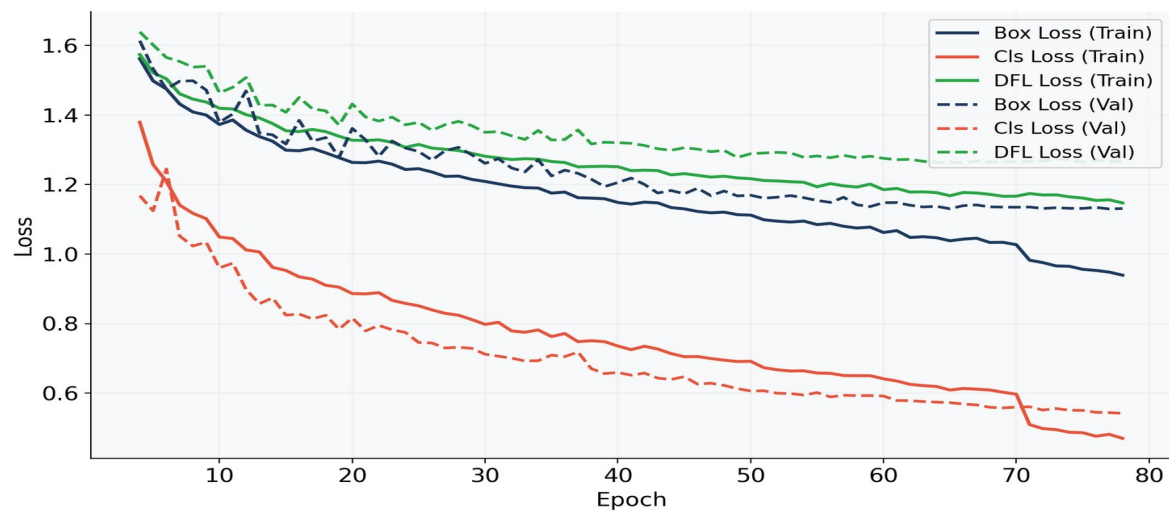
Recall

Seatbelt Model — Training Curves

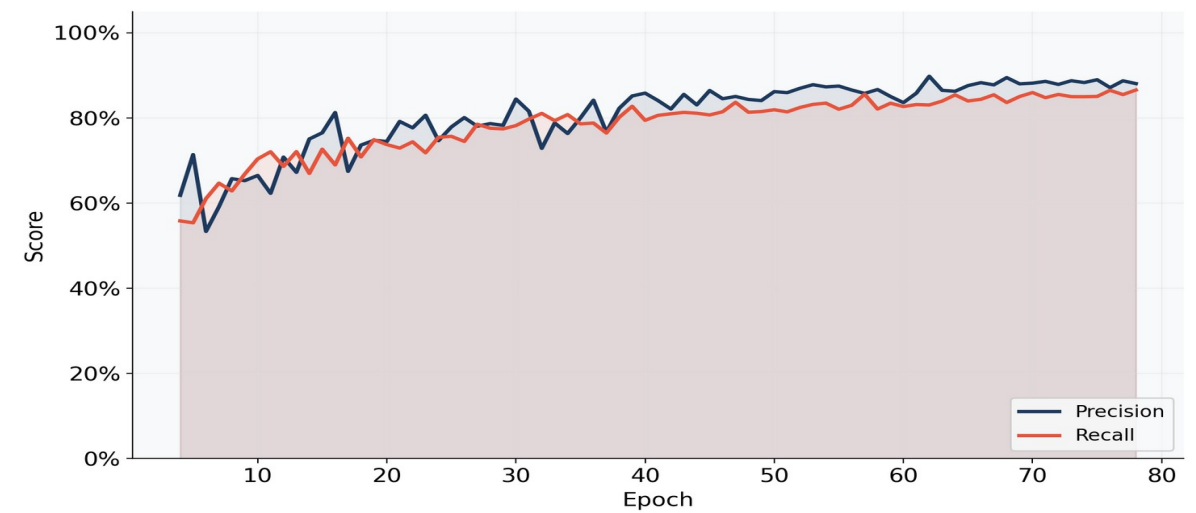


Seatbelt Model — Detailed Metrics

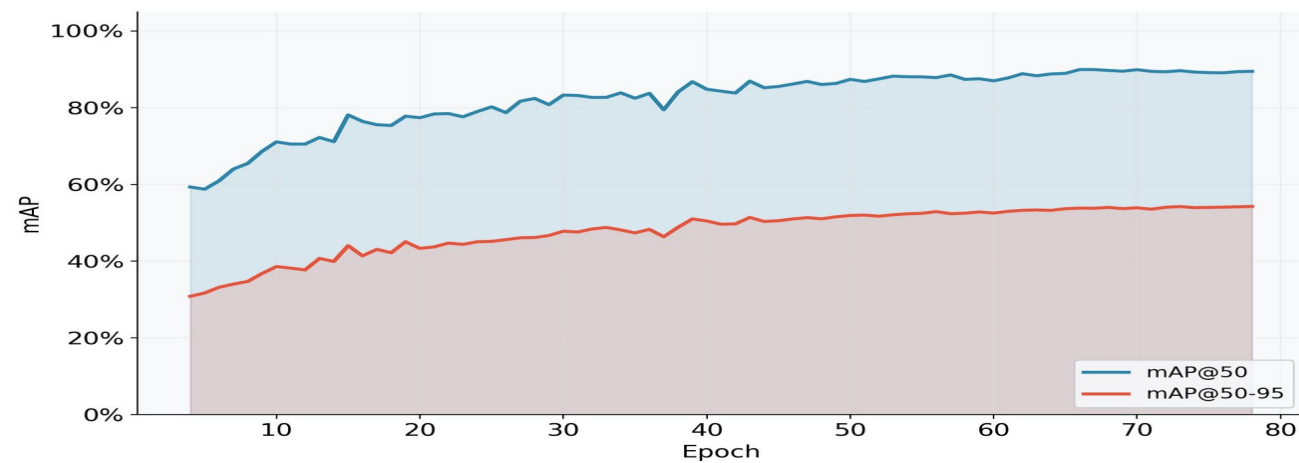
Seatbelt+Mobile (YOLOv11m) — Training & Validation Loss



Seatbelt+Mobile (YOLOv11m) — Precision & Recall



Seatbelt+Mobile (YOLOv11m) — mAP Score



07

Car Detection

YOLO26s — Vehicle detection & tracking

Vehicle Detection & Tracking

YOLO26s Car Detector

Architecture	YOLO26s (small — fast)
Input Size	640 × 640
Detection Classes	car (2), bus (5), truck (7)
Tracking	ByteTrack for video frame tracking
Confidence	0.4 threshold
Purpose	Locate vehicles before interior/plate analysis

Video Pipeline

- 1 Scan all video frames for cars
- 2 ByteTrack assigns track IDs per pass
- 3 Deduplication: IoU > 0.5 + frame proximity < 30
- 4 Match cars by spatial proximity (NOT track ID across passes)
- 5 Find plate crops for each unique car
- 6 Rank plate candidates by crop area

08

Training Infrastructure

Kaggle T4 GPU — Training at scale

Training Infrastructure & Config

Hardware & Platform

Platform	Kaggle Notebooks (free T4 GPU)
GPU	NVIDIA Tesla T4 (16GB VRAM)
OCR V2 batch	16 (batch=32 OOMs for YOLOv11m)
Seatbelt batch	32 (fits for YOLOv11m at 640)
OCR V2 Time	$\sim 7 \text{ min/epoch} \times 100 = \sim 12 \text{ hours}$
Seatbelt Time	$\sim 4 \text{ min/epoch} \times 80 = \sim 5 \text{ hours}$

Lessons Learned

Batch Size Matters

YOLOv11m batch=32 OOMs on T4 — use batch=16

Class Weighting

Weighted BCE loss didn't improve results — normal training with more epochs was better

Augmentation

RandAugment + MixUp(0.1) gave best results

Warmup Removal

First 3 epochs are warmup — exclude from analysis charts

Image Size

Must match training imgsz=640 or predictions break

Roboflow Labels

Some exports have polygon data (>5 values/line) — truncate to first 5

09

Live Demo

Streamlit Web Interface

Streamlit Web Interface



Smart Sidebar

Real-time confidence sliders for all models (step=0.01). Model status indicators. Cache-min detection for instant filtering.



Photo Mode



Upload PNG images → detect cars as clickable cards → interior analysis (seatbelt/mobile) → plate OCR with enhancement comparison.



Clickable Car Grid



Detected vehicles shown as image cards with green/red plate badge. Click to analyze. Filtered by confidence slider in real-time.



9-Way OCR Comparison



3 OCR models × 3 enhancement levels = 9 results per plate. Annotated images with per-class colored boxes and Franco labels.



Sample Gallery



Built-in sample images gallery with compact expandable grid. Click any thumbnail to process.



Violation Badges



Colored Streamlit badges: red 'No Belt', green 'Belt', red 'Phone'. Instant visual feedback for safety violations.

Key Results

99.2%

OCR V2
mAP@50

90.0%

Seatbelt
mAP@50

9

OCR Results
Per Plate

6

Specialized
Models

38

Character
Classes

<1s

Per Image
Inference

Challenges We Faced

Video Tracking Unreliability

ByteTrack assigns fresh track IDs on each video pass. Cross-pass car matching requires spatial proximity (IoU), not track IDs. Video mode was removed pending a better approach.

Class Imbalance in OCR

Some Arabic characters appear far more frequently than others. Attempted weighted BCE loss — but normal training with more epochs performed better.

LapSRN Instability

Super-resolution model crashes on certain image shapes (cv2.error in merge). Required try/except fallback to original image.

GPU Memory Constraints

Kaggle T4 has 16GB VRAM. YOLOv11m batch=32 OOMs — had to use batch=16. This doubles training time.

Roboflow Label Format

Some Roboflow YOLO exports include polygon data (>5 values per line). Had to truncate to standard format (class x y w h).

Arabic Text Rendering

Bidirectional text (numbers LTR, Arabic RTL) requires careful handling. Used `separate_chars()` to split and reverse correctly.

What's Next?



Video Mode Revival

High

Revisit video processing with better tracking approach. Current plate-first method was too slow.



Auto Mode

High

Automatically select best plate crop using sharpness scoring, not just area ranking.



Color Classification

Medium

K-means clustering for vehicle color detection. Config ready (COLOR_KMEANS_CLUSTERS), implementation pending.



Speed Estimation

Medium

Use car tracking between frames to estimate vehicle speed from video feeds.



Improved Deduplication

Medium

Current IoU-based dedup works, but visual similarity (feature matching) would be more robust.



Model Optimization

Low

Quantization and TensorRT for faster inference. Deploy on edge devices (Jetson, Coral).

Thank You

Questions?

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